

Personal Portfolio Application

Comprehensive Project Overview & Detailed File-by-File Technical Documentation

Developer: Abhijeet Vilayate	Tech Stack: React 19, Vite, Bootstrap 5, EmailJS
Role: Full Stack Developer	Routing: React Router DOM (HashRouter)
Education: B.E. Chemical Engineering (AISSMS COE)	Deployment Target: GitHub Pages (gh-pages)
Project Location: New-Portfolio	Total Files Documented: 55+ Files (Root, Public, Src, Scaffold)

1. Executive Project Overview

This project represents the personal developer portfolio website for **Abhijeet Vilayate**, a passionate Full Stack Developer with expertise in modern web technologies, React.js, Java, Spring Boot, RESTful APIs, and relational databases. Built from the ground up using **React 19** and standard **Vite**, the application combines a responsive, aesthetically rich UI with component-driven architecture.

The project is designed to deliver a seamless user experience across mobile, tablet, and desktop devices, leveraging **Bootstrap 5**, **Bootstrap Icons**, custom CSS grid/flexbox layouts, animated background shapes, SVG-based circular progress gauges, interactive filter pills, and direct email communication capabilities powered by the **EmailJS** SDK.

Key Architectural & Design Highlights:

- **Single Page Application (SPA) Routing:** Utilizes React Router DOM configured with HashRouter to ensure smooth client-side navigation without requiring server-side rewrite rules when hosted on static platforms like GitHub Pages.
- **Modular Component Architecture:** Strictly decouples application concern into shared layout components (Header, Footer, StatsBanner) and dynamic page views (Home, About, Skills, Projects, Contact).
- **Interactive Data Visualization:** Custom SVG circular gauges dynamically calculate strokeDashoffset values to render SVG progress rings for technical skills.
- **Dynamic Multi-Category Filtering:** Stateful pill buttons on the Projects page allow visitors to filter featured projects across Web Development, Java/JDBC, Frontend, and Full Stack categories in real-time.
- **Live Contact Form Integration:** Integrates @emailjs/browser to enable direct email dispatch from website visitors to the developer's inbox without requiring a backend web server.
- **Future-Proof Architecture (ERP Module Scaffold):** Includes a pre-structured scaffold directory (src/src/) comprising 29 sub-module files prepared for expanding the portfolio with an embedded Enterprise Resource Planning (ERP) & College Management System.

2. Repository Structure & File Statistics

Directory Category	File Count	Primary Technologies & Description
Root Level Configuration	7 Files	Vite config, package manifests, ESLint, HTML entry point, Git setup
Public Directory (public/)	5 Files	Static brand logo PNG, SVG favicons, downloadable resume PDFs

Application Core (src/)	4 Files	React root script (main.jsx), router layout (App.jsx), base CSS
Main Shared Components (src/Main/ & src/)	5 Files	Sticky Header, Footer, StatsBanner with matching CSS modules
Page Components (src/Pages/)	10 Files	Home, About, Skills, Projects, Contact pages & dedicated CSS stylesheets
Assets Directory (src/assets/)	9 Files	Developer portrait cutouts, project screenshot assets, SVG logos, PDF resume
ERP Module Scaffold (src/src/)	29 Files	Scaffold architecture files for ERP/Management System (routes, context, services, views)
TOTAL FILE COUNT	59 Files	Complete Portfolio & Architecture Workspace

3. Detailed File-by-File Technical Breakdown

Group A: Root Configuration & Build Files

package.json	JSON
Primary Purpose: Project manifest containing scripts, production dependencies, and dev tooling. Defines project metadata (portfolio, version 0.0.0, ESM module type). Contains NPM scripts: dev (starts Vite server), build (compiles bundle), lint (ESLint check), preview (previews build), predeploy, and deploy (deploys dist folder to GitHub Pages using gh-pages). Lists core dependencies: @emailjs/browser (v4.4.1), bootstrap (v5.3.8), bootstrap-icons (v1.13.1), react (v19.2.7), react-bootstrap (v2.10.10), react-dom (v19.2.7), react-icons (v5.7.0), and react-router-dom (v7.18.1).	
vite.config.js	JavaScript
Primary Purpose: Vite bundler and development server configuration. Exports default Vite configuration utilizing @vitejs/plugin-react for Fast Refresh, JSX transformation, and optimized ESM bundling.	
index.html	HTML
Primary Purpose: Main HTML template and document entry point. Defines HTML5 document metadata, sets the browser title to 'Abhijeet Vilayate Full Stack Developer', links the site favicon (/av-logo.png), configures the mobile viewport, and provides the <div id='root'></div> mounting point for React rendering.	
eslint.config.js	JavaScript
Primary Purpose: ESLint flat configuration file for code quality and linting. Defines code linting rules, enables React hooks and React refresh plugins, sets browser globals, and specifies linting rules across JS/JSX source files.	
README.md	Documentation
Primary Purpose: Markdown documentation for the repository. Contains default Vite + React template notes, React compiler advice, ESLint setup instructions, and repository identifier heading.	
.gitignore	Config
Primary Purpose: Version control exclusion rules. Specifies paths and patterns ignored by Git, including node_modules/, build output dist/, local environment files, and IDE cache folders.	

package-lock.json	JSON
Primary Purpose: NPM dependency lockfile. Automatically generated lockfile recording exact version dependency trees, integrity hashes, and package resolutions for deterministic npm builds.	

Group B: Public Directory Static Assets (public/)

public/av-logo.png	PNG
Primary Purpose: Primary brand logo image asset. Personal developer logo PNG displayed in the website sticky navbar header and global footer components.	
public/Abhijeet_Vilayate_Resume.pdf	PDF
Primary Purpose: Downloadable developer resume file. Official PDF resume document of Abhijeet Vilayate made directly available for public download via navbar and contact buttons.	
public/resume.pdf	PDF
Primary Purpose: Alias copy of developer resume. Alternative standardized static link filename for accessing the developer's resume PDF.	
public/favicon.svg & public/icons.svg	SVG
Primary Purpose: Browser tab icon and vector icon sprite library. High-definition SVG vector favicon file for browser address bars and tab header icons, alongside SVG icon definitions.	

Group C: Application Core Files (src/)

src/main.jsx	React JSX
Primary Purpose: React application initialization & bootstrapper script. Imports React StrictMode, createRoot from react-dom/client, Bootstrap CSS (bootstrap.min.css), Bootstrap JavaScript bundle (bootstrap.bundle.min.js), Bootstrap Icons CSS (bootstrap-icons.css), global stylesheet index.css, and mounts the root <App /> component into the DOM element #root.	
src/App.jsx	React JSX
Primary Purpose: Main routing component establishing application structure. Configures HashRouter (as Router) for seamless static hosting compatibility. Renders the shared global Header component above a Routes container mapping 5 page views: / (Home), /about (About), /skills (Skills), /projects (Projects), and /contact (Contact).	
src/index.css & src/App.css	CSS Stylesheet
Primary Purpose: Global root stylesheets for typography, resets, and layout defaults. Provides foundational styling rules, CSS variable declarations, font family overrides, and global application container reset rules.	

Group D: Shared Layout Components (src/Main/ & src/)

src/Main/Header.jsx	React JSX
Primary Purpose: Sticky global top navigation bar component. Uses useLocation() from React Router to apply dynamic active CSS classes to navigation links. Displays the brand logo (av-logo.png), developer name, title, collapsing Bootstrap navbar toggler for mobile devices, and a 'Download Resume' CTA button pointing to Abhijeet_Vilayate_Resume.pdf.	
src/Main/Header.css	CSS Stylesheet
Primary Purpose: Styles for sticky navbar, brand logo, and navigation buttons. Defines backdrop glassmorphism styling (backdrop-filter: blur), sticky positioning, link hover state indicators, purple color highlights, and responsive mobile nav drawer adjustments.	
src/Footer.jsx	React JSX
Primary Purpose: Global website footer component. Features a 3-column layout displaying personal bio summary, brand logo, quick navigation links (Home, About, Skills, Projects, Experience, Contact), direct email link, social media icon buttons (GitHub, LinkedIn, Instagram, Twitter/X), dynamic current copyright year generator, and a smooth scroll-to-top button (BsArrowUpCircleFill).	
src/Footer.css	CSS Stylesheet
Primary Purpose: Styles for global footer layout, social icons, and top button. Configures dark theme card backgrounds, link transition hover effects, social media icon pill buttons, and responsive grid column alignment.	
src/Main/StatsBanner.jsx	React JSX
Primary Purpose: Key performance indicators & statistics banner component. Renders a responsive 4-column metric card showcasing portfolio metrics: 3+ Projects Completed (bi-code-slash), 4+ Technologies Used (bi-layers-fill), 100% Passion & Dedication (bi-trophy-fill), and Continuous Learning Journey (bi-rocket-takeoff-fill).	
src/Main/StatsBanner.css	CSS Stylesheet
Primary Purpose: Styles for stats banner container, icon circles, and metric titles. Custom styling for shadow elevations, icon background circles, bold metric numbers, and muted subtitle text.	

Group E: Dynamic Page Components (src/Pages/)

src/Pages/Home.jsx	React JSX
Primary Purpose: Landing page hero view and personal overview. Renders the top hero section featuring greeting wave badge, developer title ('Abhijeet Vilayate - Full Stack Developer'), bio description, action buttons ('Hire Me' linking to /contact, 'View My Work' linking to /projects), social media profile links (GitHub, LinkedIn, Email, Twitter/X), organic background blob backdrop, developer portrait image (ME.png), and embedded StatsBanner.	
src/Pages/Home.css	CSS Stylesheet

Primary Purpose: Home page hero styling, animated blobs, and custom button themes.

Defines hero background glowing backdrop, organic image cutout shapes, purple button styling (btn-purple, btn-outline-purple), and social button hover animations.

src/Pages/About.jsx

React JSX

Primary Purpose: Detailed developer bio, background, education, and tech skills grid.

Presents developer portrait with floating 'Open to Work' badge, 6-item personal information grid (DOB: 26 Dec 2002, Education: B.E. Chemical Eng, Email, Phone, Location: Pune, Freelance status), and 4 detailed cards: **My Journey**, **Education Timeline** (2022-2026 B.E. AISSMS COE, 2019-2021 HSC, 2018-2019 SSC), **What I Do** (Frontend, Backend, APIs, DB, Clean Code), and **Technologies I Love** (HTML5, CSS3, JS, React, Java, Spring Boot, MySQL, Git).

src/Pages/About.css

CSS Stylesheet

Primary Purpose: Styles for portrait backdrop glow, timeline dot connectors, and info cards.

Custom CSS for portrait glowing card backdrop, floating badge positioning, info box icons, vertical education timeline dots and connecting lines, and tech icon boxes.

src/Pages/Skills.jsx

React JSX

Primary Purpose: Interactive technical skills & proficiency visualization page.

Renders page header, top circular SVG progress gauges calculating SVG strokeDashoffset rings for 7 key skills (HTML 95%, CSS 90%, JS 90%, Bootstrap 85%, React+Vite 90%, Spring Boot 85%, Python 90%), 8 category cards (Frontend, Backend, Programming & ML, Database & Cloud, Tools & Platforms, Other Skills, Soft Skills, Learning & Growth), and sub-component SkillBar rendering animated progress bars.

src/Pages/Skills.css

CSS Stylesheet

Primary Purpose: Custom SVG gauge math styles, progress bars, and category icon boxes.

Defines SVG circle viewBox dimensions, stroke dasharray calculations, category card icons, progress bar track backgrounds, and rounded pill bar fills.

src/Pages/Projects.jsx

React JSX

Primary Purpose: Filterable portfolio showcase page displaying featured software projects.

Features stateful category filter pills ('All Projects', 'Web Development', 'Java / JDBC', 'Frontend', 'Full Stack') filtering 3 projects: 1. **ERP College Management System**: Java Servlets, React 19, Oracle SQL, BCrypt, PL/SQL, RBAC across 8 roles. 2. **Netflix Home Page Clone**: HTML5, CSS3 flexbox/grid responsive clone. 3. **Personal Portfolio Website**: React.js, Vite, Bootstrap 5. Each card includes preview images, technology tags, bullet points, GitHub links, and live demo buttons.

src/Pages/Projects.css

CSS Stylesheet

Primary Purpose: Styles for filter pills, project cards, hover scaling, and action buttons.

Category pill active states (purple background), project card hover transform effects, image preview wrapper cropping, and outline button styles.

src/Pages/Contact.jsx

React JSX

Primary Purpose: Interactive contact page with EmailJS integration.

Integrated with @emailjs/browser SDK. Renders hero graphic, contact info card (Email, Phone, Location, Availability, Social Buttons), interactive contact form (Name, Email, Subject, Message) with loading spinner state dispatching emails via EmailJS (service_gxqg5sa, template_mbm3nii), 4 'Why Work With Me?' cards (Quick Response, Full Stack, Modern Tech, Collaborative), CTA callout banner, and page footer.

src/Pages/Contact.css	CSS Stylesheet
Primary Purpose: Styles for contact hero, custom form inputs, why-cards, and CTA banner. Custom form input focus borders, purple submit button, glowing blob backgrounds, card hover transitions, gradient CTA banner, and bottom note strip.	

Group F: Component Image & Media Assets (src/assets/)

src/assets/ME.png	PNG Image
Primary Purpose: Developer portrait cutout photo. High quality transparent PNG cutout of developer Abhijeet Vilayate used as primary profile photo on Home and About page hero cards.	
src/assets/Management.png	PNG Image
Primary Purpose: ERP College Management System preview image. Project showcase screenshot image displaying the dashboard UI of the ERP College Management System project.	
src/assets/Netflix-clone.png	PNG Image
Primary Purpose: Netflix landing page clone preview image. Project showcase screenshot image displaying the Netflix clone landing page UI.	
src/assets/New portfolio.png	PNG Image
Primary Purpose: Personal Portfolio website screenshot. Project showcase screenshot displaying the Personal Portfolio web application UI.	
src/assets/av-logo.png & hero.png	PNG Image
Primary Purpose: Brand logo asset and hero illustration image. Primary brand icon PNG imported across React navbar/footer components, and supporting hero graphic PNG asset.	
src/assets/react.svg & vite.svg	SVG Image
Primary Purpose: Official React and Vite framework logo graphics. Vector SVG logos rendered in 'Technologies I Love' tech stack grids.	

Group G: ERP / Management System Module Scaffold (src/src/)

The src/src/ directory contains a pre-structured scaffold comprising **29 architectural placeholder files** designed for future expansion of the portfolio with an integrated Enterprise Resource Planning (ERP) & College Management System module. The scaffold includes clear architectural layering across assets, common UI components, modal dialogs, context state providers, router hooks, service APIs, and feature views.

src/src/assets/index.css & theme.css	Scaffold
Primary Purpose: ERP Dashboard Scaffold File Theme and base style blueprints for the ERP dashboard module.	

src/src/components/common/DataTable.jsx	Scaffold
Primary Purpose: ERP Dashboard Scaffold File Reusable data grid table component for managing student/faculty lists.	
src/src/components/common/Navbar.jsx & Sidebar.jsx	Scaffold
Primary Purpose: ERP Dashboard Scaffold File Dashboard top navigation bar and side drawer navigation components.	
src/src/components/common/StatCard.jsx & Footer.jsx	Scaffold
Primary Purpose: ERP Dashboard Scaffold File Dashboard summary metric cards and internal footer components.	
src/src/components/modals/ApplyLeaveModal.jsx	Scaffold
Primary Purpose: ERP Dashboard Scaffold File Modal dialog form for student/employee leave requests.	
src/src/components/modals/IssueBookModal.jsx	Scaffold
Primary Purpose: ERP Dashboard Scaffold File Modal dialog form for library book issuing operations.	
src/src/components/modals/LogoutModal.jsx	Scaffold
Primary Purpose: ERP Dashboard Scaffold File Confirmation modal dialog for user session logout.	
src/src/components/public/HeroSection.jsx, ContactForm.jsx, ServicesGrid.jsx	Scaffold
Primary Purpose: ERP Dashboard Scaffold File Public ERP portal landing components.	
src/src/context/AuthContext.jsx	Scaffold
Primary Purpose: ERP Dashboard Scaffold File React Context Provider for handling ERP user authentication state.	
src/src/routes/AppRoutes.jsx & ProtectedRoute.jsx	Scaffold
Primary Purpose: ERP Dashboard Scaffold File Role-Based Access Control (RBAC) route definitions and guard wrappers.	
src/src/services/api.js, authService.js, attendanceService.js, feeService.js	Scaffold
Primary Purpose: ERP Dashboard Scaffold File REST API integration services for login, attendance, fees, and CRUD operations.	
src/src/shared-modules/AssignmentsView.jsx, AttendanceView.jsx, FeeStatusView.jsx	Scaffold

Primary Purpose: ERP Dashboard Scaffold File Feature view modules for managing assignments, student attendance, and fee tracking.	
src/src/shared-modules/LeaveRequestsView.jsx, LibraryView.jsx, NoticesView.jsx	Scaffold
Primary Purpose: ERP Dashboard Scaffold File Feature view modules for leave approvals, digital library catalog, and notice announcements.	
src/src/shared-modules/ProfileView.jsx, SubjectsView.jsx, TimetableView.jsx	Scaffold
Primary Purpose: ERP Dashboard Scaffold File User profile management, course subject mapping, and class timetable schedule views.	

4. Key NPM Dependencies & Tooling Matrix

Package Name	Version	Category	Functionality in Project
react	^19.2.7	Core Library	Core UI view library powering component rendering and hooks.
react-dom	^19.2.7	Core DOM	DOM rendering package for mounting React components to index.html.
react-router-dom	^7.18.1	Routing	Client-side routing engine using HashRouter for single-page navigation.
bootstrap	^5.3.8	UI Framework	Responsive grid layout system, utility classes, and components.
bootstrap-icons	^1.13.1	Icon Library	Font icon library for buttons, skill bars, and section headers.
react-icons	^5.7.0	Icon Components	React icon components for GitHub, LinkedIn, Instagram, and Twitter X.
@emailjs/browser	^4.4.1	API SDK	Client-side SDK for dispatching contact form emails directly.
vite	^8.1.1	Dev Tooling	Next-gen frontend bundler and development server with HMR.
gh-pages	^6.3.0	Deployment	Automation tool for publishing compiled production builds to GitHub Pages.

5. Build, Execution & Deployment Guide

The project includes pre-configured standard scripts in package.json to facilitate development, linting, testing, building, and seamless automated deployment to **GitHub Pages**.

Standard Terminal Commands:

- npm run dev - Starts the local Vite development server with Hot Module Replacement (HMR).
- npm run build - Compiles and optimizes production assets into the static dist/ directory.
- npm run preview - Starts a local web server to preview the production bundle build.
- npm run lint - Runs ESLint code analysis across all JavaScript and React JSX files.
- npm run deploy - Automatically executes npm run build and deploys the dist/ directory to the gh-pages branch.